

System Requirements Assessment — Current VM

Date: 2026-04-01

Machine: DeepAgent Cloud VM (AWS)

Hardware Assessment

Component	Available	UE5 Requirement	Status
CPU	32 cores (Intel Xeon 6975P-C)	6+ cores	✓ Exceeds
RAM	247 GB	32 GB	✓ Exceeds
Disk	335 GB free / 503 GB total	200 GB	✓ Sufficient
GPU	None	NVIDIA GTX 1080+	✗ Missing
OS	Debian 12 (bookworm) x86_64	Ubuntu 22.04+	✓ Compatible

Limitation Impact

What CAN be done on this VM:

- ✓ Project configuration and file setup
- ✓ Script creation and preparation
- ✓ iOS app code updates (Config.swift, PixelStreamingService)
- ✓ UE5 source download and partial build (build tools only)
- ✓ Asset manifest and import script generation
- ✓ Docker/deployment configuration

What CANNOT be done on this VM:

- ✗ UE5 Editor rendering (requires GPU)
- ✗ Asset import into UE5 (requires running Editor)
- ✗ Project cooking/packaging (requires GPU for shader compilation)
- ✗ Pixel Streaming server (requires GPU for real-time rendering)
- ✗ iOS build (requires macOS + Xcode)

Recommended GPU Machines

Cloud (for Linux server build + cooking)

- **AWS:** g5.4xlarge (NVIDIA A10G, 64GB RAM) — ~\$1.62/hr

- **GCP:** n1-standard-8 + T4 GPU — ~\$0.95/hr
- **Azure:** NC6s_v3 (NVIDIA V100) — ~\$3.06/hr

Local (for full development)

- **macOS:** Mac Studio M2 Ultra (for iOS + UE5 Editor)
- **Windows/Linux:** RTX 3080+ desktop (for UE5 Editor + cooking)

Transfer Instructions

```
# Tar the project for transfer
cd /home/ubuntu
tar czf fel-project.tar.gz rork-final-evolution-lab/

# SCP to GPU machine
scp fel-project.tar.gz user@gpu-machine:~/

# On GPU machine
cd ~
tar xzf fel-project.tar.gz
cd rork-final-evolution-lab
./scripts/ue5_setup/fel_complete_pipeline.sh
```